# A Trace-Class Bergman Fredholm Product for an Order-Sensitive Möbius Iterated System

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**Abstract**

We freeze the nonlinear branches $\phi_a(z) = 1/(a+z)$ for $a \in \{3,6\}$ on the unit disk. Their closed images are strictly separated, and the unweighted composition sum on normalized Bergman space is trace class with an explicit norm bound. Every word has a unique fixed point, a quadratic-algebraic multiplier read from an integer Möbius matrix, and an exact composition trace. These identities give an all-period trace formula and a primitive Fredholm product. Two non-cyclic same-count words have different multipliers and traces, repairing common-linear affine location blindness by intrinsic nonlinear order. The construction is dynamical and does not supply arithmetic Euler factors or a target spectral match.

## 1 Progress over the prior gate

For a common-linear affine alphabet, determinant weights can collapse a word to symbol populations: rearranging positions while preserving counts then changes no weight. Here matrix multiplication is noncommutative and the branch images are geometrically separated. The source itself therefore retains cyclic order; no external phase label is appended.

## 2 Frozen disk geometry and trace class

Let $\mathbb{D} = \{z : |z| < 1\}$ and let $A^2(\mathbb{D})$ use normalized area measure, so $e_n(z) = \sqrt{n+1}z^n$ is orthonormal. Define

$$\phi_a(z) = \frac{1}{a+z} \quad (a = 3, 6), \qquad \mathcal{L} = C_{\phi_3} + C_{\phi_6}, \qquad C_\phi f = f \circ \phi.$$

The image of the closed disk under $\phi_a$ is the disk

$$\overline{\mathbb{D}}\left(\frac{a}{a^2-1}, \frac{1}{a^2-1}\right).$$

Thus the respective centers and radii are $(3/8, 1/8)$ and $(6/35, 1/35)$; the largest image moduli are $1/2$ and $1/5$, and the closed-image gap is

$$\frac{3}{8} - \frac{1}{8} - \frac{6}{35} - \frac{1}{35} = \frac{1}{20}.$$

Moreover $\sup_{\mathbb{D}} |\phi_a'| \le (a-1)^{-2}$, giving $1/4$ and $1/25$.

**Theorem 1** (trace-class owner)**.** *The operator $\mathcal{L}$ is trace class on $A^2(\mathbb{D})$ and $\|\mathcal{L}\|_1 \le 89/16$.*

*Proof.* The Bergman expansion gives the rank-one representation

$$C_{\phi_a} f = \sum_{n\geq 0} \langle f, e_n \rangle \sqrt{n+1}\, \phi_a^n.$$

Each coefficient functional has norm one. If $\sup|\phi_a| \leq r_a$, then $\|\sqrt{n+1}\phi_a^n\|_{A^2} \leq \sqrt{n+1} r_a^n$. Hence the nuclear series converges; the deliberately coarse inequality $\sqrt{n+1} \leq n+1$ gives

$$\|\mathcal{L}\|_1 \leq \sum_{n\geq 0}(n+1)2^{-n} + \sum_{n\geq 0}(n+1)5^{-n} = 4 + \frac{25}{16} = \frac{89}{16}.$$

□

## 3 All-word fixed points and traces

Put $M_a = \left(\begin{smallmatrix} 0 & 1 \\ 1 & a \end{smallmatrix}\right)$. For $w = a_1 \cdots a_n$, let

$$\Phi_w = \phi_{a_1} \circ \cdots \circ \phi_{a_n}, \qquad M_w = M_{a_1} \cdots M_{a_n} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

This matrix convention represents the displayed function composition. Composition operators reverse it: $C_{\phi_{a_1}} \cdots C_{\phi_{a_n}} = C_{\phi_{a_n} \circ \cdots \circ \phi_{a_1}}$. Reversal is a bijection on all length-$n$ words and will be relabelled only when summing them. Each word is a strict contraction of the closed disk and therefore has one fixed point in $\mathbb{D}$.

**Theorem 2** (quadratic word trace)**.** *Let $t = \operatorname{tr} M_w$, $\delta = \det M_w = (-1)^n$, and $\Delta = t^2 - 4\delta$. Then*

$$z_w = \frac{A - D + \sqrt{\Delta}}{2C}, \qquad \lambda_w = \Phi_w'(z_w) = \frac{t - \sqrt{\Delta}}{t + \sqrt{\Delta}},$$

*and*

$$\operatorname{Tr} C_{\Phi_w} = \frac{1}{1-\lambda_w} = \frac{1}{2} + \frac{t}{2\sqrt{\Delta}}.$$

*Proof.* The fixed equation is $Cz^2 + (D-A)z - B = 0$, whose discriminant is $(D-A)^2 + 4BC = t^2 - 4\delta$. Contraction selects the displayed root in $\mathbb{D}$. Also $Cz_w + D = (t + \sqrt{\Delta})/2$ and $\Phi_w'(z_w) = \delta/(Cz_w + D)^2$, which gives the multiplier.

Conjugate $z_w$ to zero by a disk automorphism. The induced composition operator is bounded and invertible on Bergman space. The conjugated symbol has form $\lambda_w z + O(z^2)$, so its monomial matrix is triangular with diagonal $1, \lambda_w, \lambda_w^2, \ldots$. The word operator is trace class: it is a product of a trace-class branch operator with bounded composition operators. Trace invariance under bounded similarity now gives the stated geometric sum. □

## 4 All periods and primitive product

Expanding $\mathcal{L}^n$ and applying Theorem 2 yields, for every $n \geq 1$,

$$\operatorname{Tr} \mathcal{L}^n = \sum_{|w|=n} \frac{1}{1-\lambda_w}. \tag{1}$$

Here equation (1) uses the reversal bijection just stated; it is not a termwise identification of the two order conventions. For primitive cyclic words $[p]$, regrouping the Fredholm logarithm gives

$$\det(I - z\mathcal{L}) = \prod_{[p]} \prod_{k\geq 0} (1 - z^{|p|}\lambda_p^k). \tag{2}$$

The factor is well defined on $[p]$: cyclic rotation preserves the trace and determinant of the matrix product and therefore preserves $\lambda_p$. Indeed, a repetition $p^r$ has $|p|$ rooted rotations and multiplier $\lambda_p^r$, while $(1-\lambda_p^r)^{-1} = \sum_{k\geq 0} \lambda_p^{rk}$. Thus its logarithmic contribution is

$$-\sum_{r\geq 1} \frac{z^{r|p|}}{r(1-\lambda_p^r)} = \sum_{k\geq 0} \log(1 - z^{|p|}\lambda_p^k).$$

Because there are at most $2^\ell/\ell$ primitive cycles of length $\ell$ and the chain-rule contraction bound gives $|\lambda_p| \leq 4^{-\ell}$, the sum of the absolute factor arguments is dominated by a constant times $\sum_{\ell\geq 1}(2|z|)^\ell/\ell$. Hence the raw product is absolutely convergent for $|z| < 1/2$. The logarithmic regrouping is initially valid near $z = 0$; analyticity and the identity theorem extend the equality throughout that disk. The left side is entire by Theorem 1; this analytic continuation does not assert that the displayed raw factors converge outside their proved disk.

# 5 Intrinsic order control and exact receipt

The words 33366 and 33636 have three digits 3 and two digits 6, but are not cyclic rotations: the rotations of the first are $33366, 33663, 36633, 66333, 63336$. Exact multiplication gives

$$M_{33366} = \begin{pmatrix} 63 & 388 \\ 208 & 1281 \end{pmatrix}, \qquad M_{33636} = \begin{pmatrix} 60 & 379 \\ 199 & 1257 \end{pmatrix}.$$

Both determinants are $-1$, whereas their traces are 1344 and 1317. For positive $t$, the trace weight is $1/2 + t/(2\sqrt{t^2+4})$. Equality of two such weights, after squaring, would force $t_1^2(t_2^2+4) = t_2^2(t_1^2+4)$ and hence $t_1 = t_2$. Thus these distinct positive traces force distinct composition traces and, because the weight is $(1-\lambda)^{-1}$, distinct multipliers.

The receipt enumerates every rooted word through period ten:

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| rooted | 2 | 4 | 8 | 16 | 32 | 64 | 128 | 256 | 512 | 1024 |
| primitive | 2 | 1 | 2 | 3 | 6 | 9 | 18 | 30 | 56 | 99 |

Thus 2,046 word cases and 226 primitive classes are represented. An independent checker reconstructs each integer matrix and ledger digest; 2,561 separate symbolic checks, byte replay, 36 repaired-hash semantic mutations, and one stale-hash mutation test the same claims.

# 6 Strict Route-A boundary

The exact separated dynamics, trace-class owner, and all-period Fredholm identity justify

$$(\mathrm{A1\_WEAK}, \mathrm{A2\_FAIL}, \mathrm{A3\_FAIL}, \mathrm{A4\_FAIL}).$$

There is no arithmetic Euler product, prime or zero table, root number, functional equation, target divisor match, automorphy, self-adjoint lift, or Hilbert–Pólya claim. Route B is unauthorized. The active scope is

`NO_BAD_EULER_OR_ROOT_NUMBER`.

# 7 Conclusion

C132 replaces position-blind common-linear weights by an intrinsically order-sensitive nonlinear system and gives its trace-class determinant at all periods. Relating that dynamical determinant to any external target is a separate, currently unsupported problem.